

Tempered Spring



Regularization

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Structure - Instruments of Constraint



Regularization



Regularization puts freedom into tension towards a preference. It guards learning against noise and overparametrization and anchors towards sensible prior expectations but can heavily bias.



priors, smoothness, continuity, simplicity, weight decay, sparsity, L1 L2 norms, bias-variance trade-off



Grants stability and graceful generalization.



Too tight, wrong direction, and variation is bent into something irrerecognizable.



Put the spring between two target cards; interpret as compromise.



"If I increase the strength of my regularizers by an order of magnitude, what will happen?"



They tempered the spring until the pendulum oscillated between chaotic reality and simplistic beauty. Later hands wound it tighter but then discarded it, frustrated by its meaningless movement.

